🔹 1. Introduction

PC-based control system in Delta is implemented using software tools like ECNavi, where the control logic is executed through an industrial PC instead of a traditional PLC.

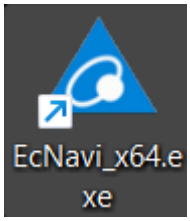
This system allows flexible configuration, high-speed communication, and advanced motion control.

🔹 2. Software Overview

The main software used is ECNavi, which can be downloaded from the Delta Electronics download center.

Using this software:

- We can **create and modify hardware configurations**
- Similar to hardware configuration in ISPSoft
- Supports system setup for motion and I/O devices

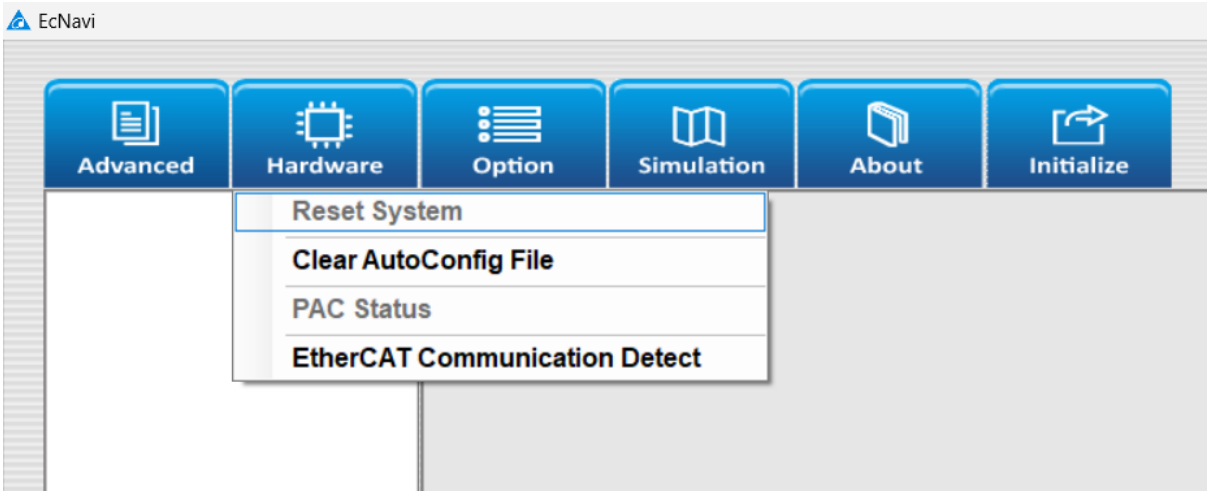


🔹 3. Hardware Configuration

Inside ECNavi:

- Hardware setup is done similar to PLC configuration
- Devices like:
  - Motion controllers
  - Servo drives
  - I/O modules
    are added and configured

👉 This is similar to how we configure modules in ISPSoft, but here it is more **PC-based and modular**



🔹 4. Library-Based Programming

One of the key features is the use of **predefined standard libraries**

- Libraries are already developed for:
  - Read/Write operations
  - Servo motion control
  - Device communication
- Instead of writing everything from scratch:
  - Engineers **reuse these libraries**
  - Just call functions and configure parameters

👉 This reduces development time and ensures reliability

🔹 5. Control Logic Implementation

In the main program:

- These libraries are used to:
  - Read inputs
  - Write outputs
  - Control servo motors
  - Communicate with other hardware devices

All machine logic is built by combining:

- Hardware configuration

- Library functions
- Custom control logic

🔹 6. Real-Time Usage in Machines

In actual machines:

- Same standard libraries are reused across projects
- Only parameters and configurations change
- This makes:
  - Development faster
  - Maintenance easier
  - System more standardized

📄 🔻 10. Limitations and Design Constraints

One of the key limitations observed in the current PC-based control system is the dependency on legacy UI technology, specifically Windows Forms (WinForms).

🔹 Existing Situation

- The current system is built using WinForms
- All internal tools, HMIs, and interfaces are developed using this framework
- Most importantly, **all standard libraries and UI components are tightly coupled with WinForms**

🔹 Reason for Continuing with WinForms

Even though modern frameworks like Windows Presentation Foundation (WPF) are available, the team continues using WinForms because:

- Existing **libraries are already developed and tested**
- These libraries directly interact with:
  - Hardware communication
  - Motion control logic
  - UI components

- Rewriting them in WPF would require:
    - Complete redesign
    - Extensive testing and validation
    - High development time and cost

👉 So, the decision is mainly based on **practical constraints, not technology limitations**

📄 🔻 **11. Future Enhancement – Migration to Modern UI Framework**

In future, the system can be upgraded to use modern UI technologies such as Windows Presentation Foundation (WPF) to improve user interface capabilities and maintainability.